

Activity Guide 1

Last updated by | Jake Kappus | Mar 30, 2023 at 11:49 AM MDT

Building a Report in Power BI Desktop

Activity Detail

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This activity walks you through creating a simple Training Tracker report with several visuals using data from an Excel file using Power BI Desktop.

Get your data

1. Open Power BI Desktop and sign in using your credentials (& CAC if necessary)

If you are already signed in (your name will appear in the upper right corner)



then continue on to Step **#2** , otherwise, follow the 🚦 Note below.

🚦 Note:

If this is your first time signing into Power BI Desktop, you will need to follow these steps.

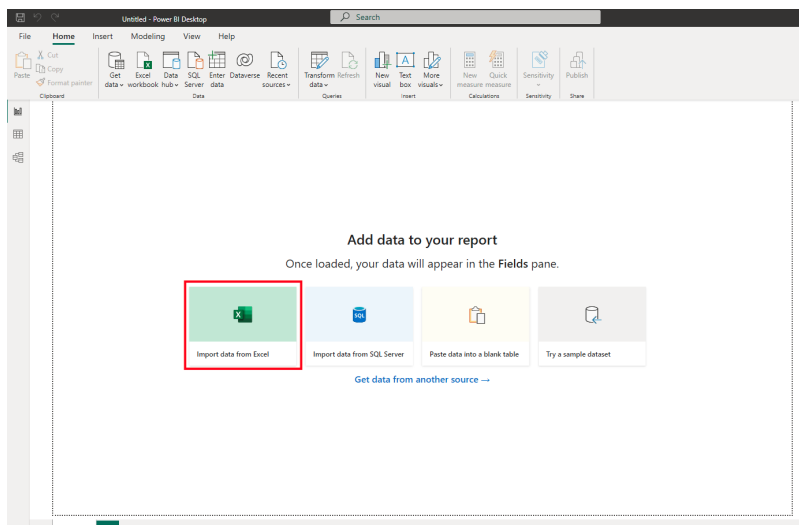
1. Under *Collaborate and share* , click on **Get Started**
2. On the *Enter your email address* screen enter your Microsoft 365 credential you would use when logging into email, Teams, etc.

You may need to restart Power BI Desktop after successfully authenticating. This is normal and configures the program to connect to the correct service in the DOD cloud.

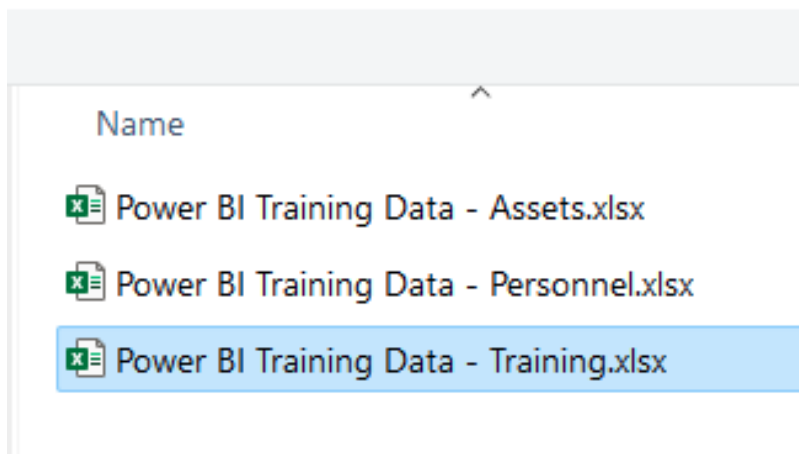
1. The first screen should be a blank report. If not, click on **File**, then **New**.

For this report, we will start with the data in an Excel file.

1. In the *Add Data to your Report* section, Click on **Import data from Excel**

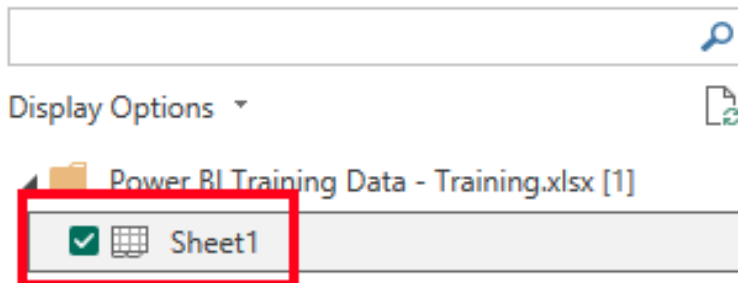


2. Choose the **Power BI Training Data - Training.xlsx** file, then click **Open**



3. On the **Training Navigator** screen, on the left side, check the box next to **Sheet 1**

Navigator



4. In the right data preview pane, scroll to the right to review the data prior to import.

Notice the column titled *Accomplished*, we want to change that to something more descriptive like **"Completed Date"**.

Click on **Transform Data** to shape this column the way we want it for our reports.

Shape the data before import

After clicking on *Transform Data*, **Power Query Editor** will open. This is a handy, yet very powerful app that will help you shape and prepare your data for use in Power BI.

1. Scroll to the right to find the **Accomplished** column. This is the column we want to change.
2. **Double Click** directly on the word **Accomplished**. This will enable editing of the column.



*Alternatively you could right click and choose **Rename***

3. Type in **Completed Date**, then the **enter** key. The column name will change.

The screenshot shows the Power Query Editor interface. The 'Query' tab is active, and the formula bar contains the M code: `= Table.TransformColumnTypes(#"Promoted Headers",{{"Person", type text}})`. The table has columns: 'mand', 'Due Date', 'Completed Date', and 'CEU Credits'. The 'Completed Date' column is highlighted, and a red box is drawn around the column header. The data rows show dates for 'Completed Date' and 'CEU Credits' values of 25.

	mand	Due Date	Completed Date	CEU Credits
1		11/6/2022	10/5/2022	25
2		11/6/2022	10/7/2022	25
3		11/6/2022	10/8/2022	25
4		11/6/2022	10/6/2022	25
5		11/6/2022	10/7/2022	25



Extra Credit

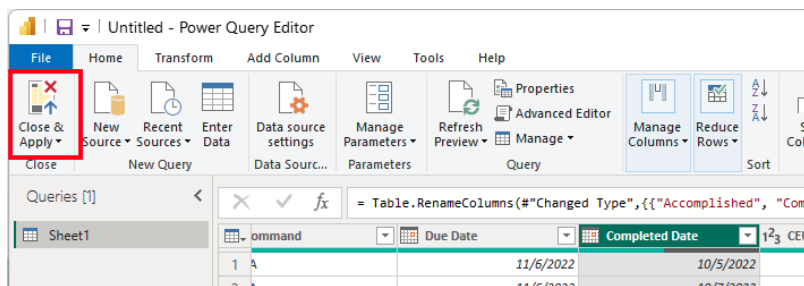
Notice in the right sidebar, under *Query Settings*, the **Applied Steps** section. These are the steps the *Power Query Editor* will take each time this report connects to the data source.

You can see there are a few steps already there. You can select each step to see its effect in the Power Query Editor. Our change specifically added the *"Changed Type"* and *"Renamed Columns"* steps..

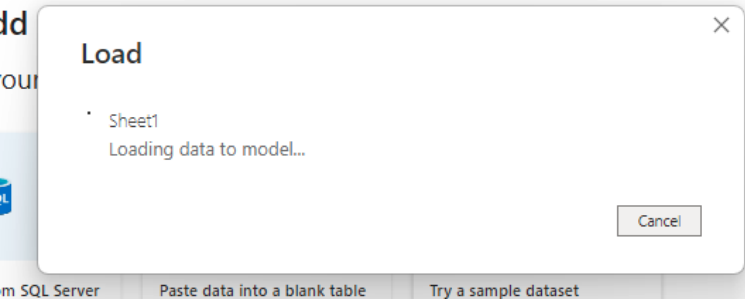
Ours is a simple example, but there may be times you may want to have changes applied to your data in a certain order (formatting numbers, or calculations).

This is how Power BI is able to shape data and how it's viewed for each person individually, without modifying the source data file.

4. Click on **Close and Apply** in the upper left corner of the *Home* menu.



Your shaped data will now be imported into Power BI Desktop.

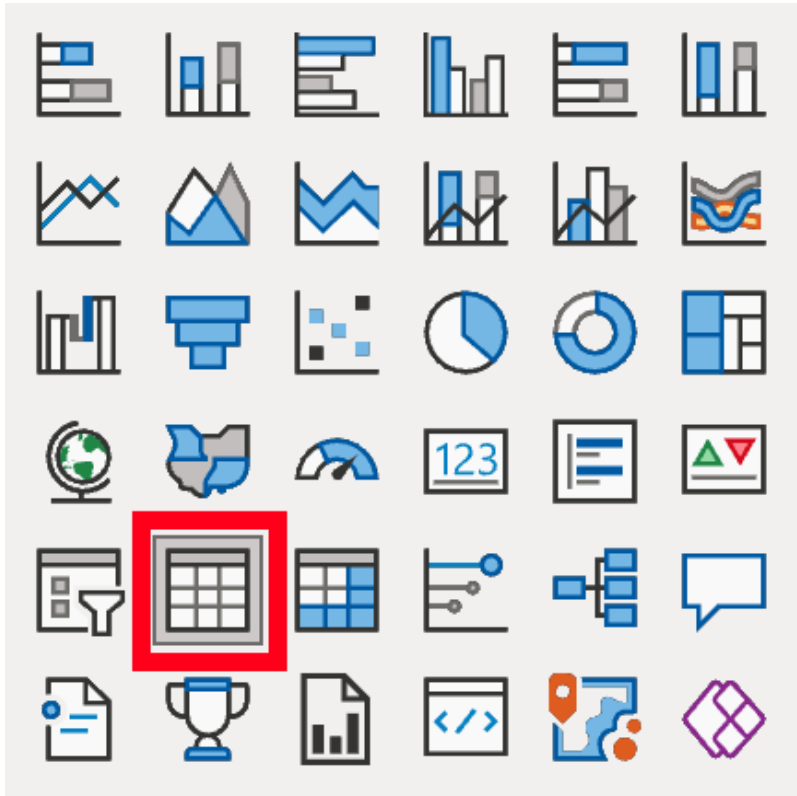


Layout your visuals: Table

If you've already done *Activity 1 - Building a Report in the Power BI Service*, the steps from this point on are almost identical to the way you would create a report in the web service. However, Power BI Desktop is much more powerful and allows for far more options than the web service can. We will try to highlight some of those in this activity.

Our first visual will be a simple table that lists the number of CEU (*Continuing Education Units*) earned in each *Command*.

1. In your blank report, under **Visualizations**, click on the Table visual



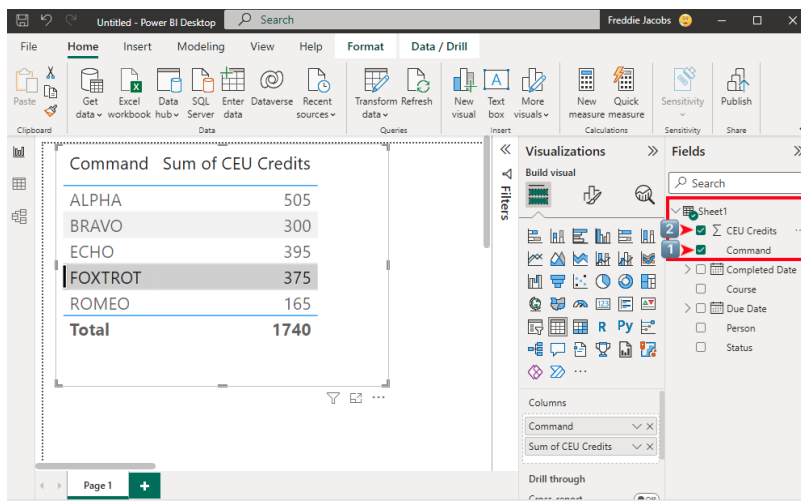
2. First we need a count of how many CEU credits have been earned in each command.

Under **Data** select the **Command** Field.

3. Now, we need to get the actual count of the CEU Credits earned in the Command. Luckily there was already a column with this data in our Excel file.

Under **Data** select the **Σ CEU Credits field**.

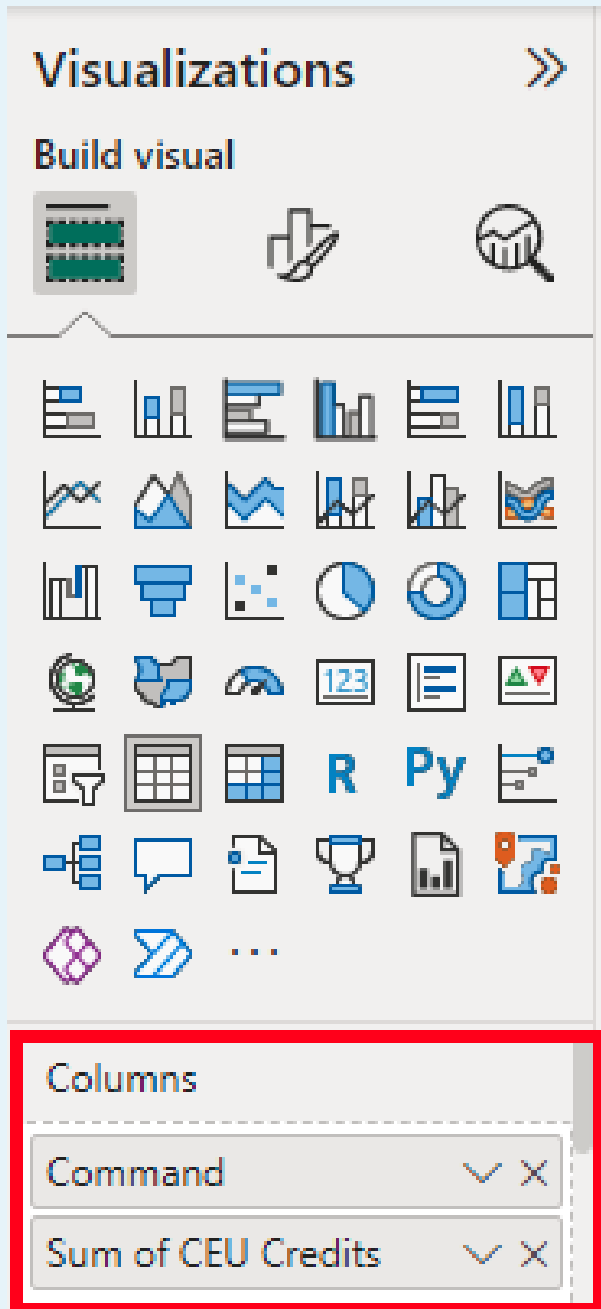
Your table will now display a count for each command.



Extra Credit

Notice the numbers in the screenshot above. With tables, it's important to the order in which you add the fields to get the right visual.

If you happened to click the fields out of order, you can change the order of the columns in the *Visualizations* pane.



Visualizations >>

Build visual

Table icon, Bar chart icon, Line chart icon

Grid of 30 visualization icons

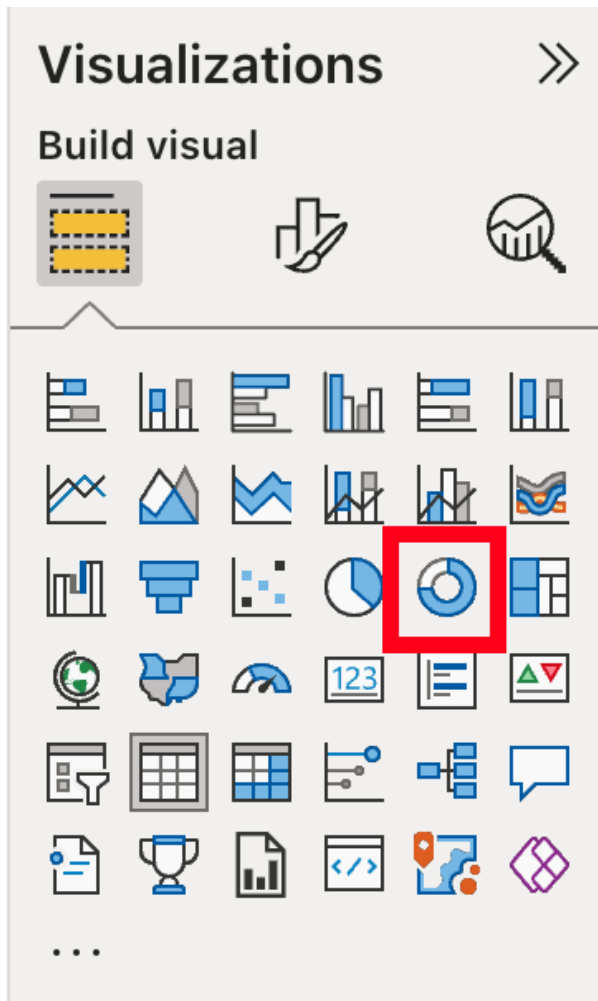
Columns

Column	Dropdown	Close
Command	▼	✕
Sum of CEU Credits	▼	✕

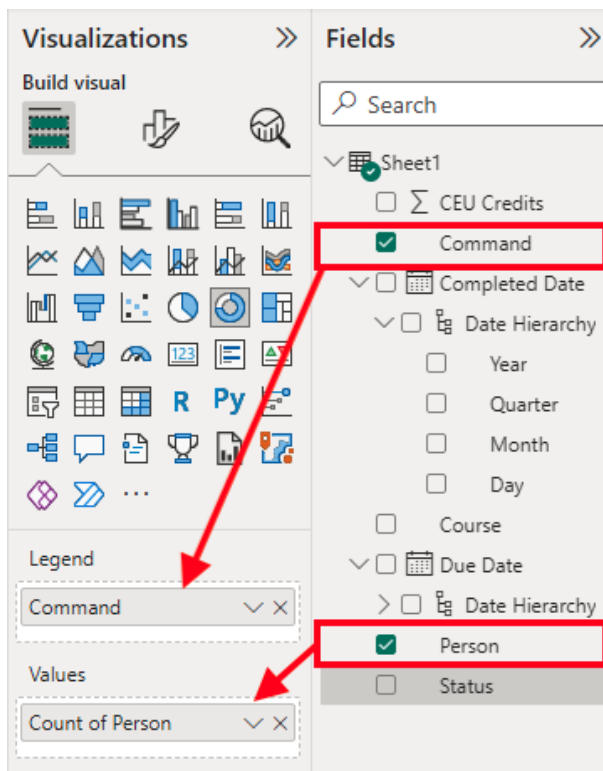
Layout your visuals: Donut chart

Tables are great, but everyone loves to see colorful charts that make it easy to interpret data quickly. Let's add a Donut chart that shows the completion status of the Security + training.

1. Click on a blank area of the report
2. In the *Visualizations* pane, select the **Donut chart** visual.



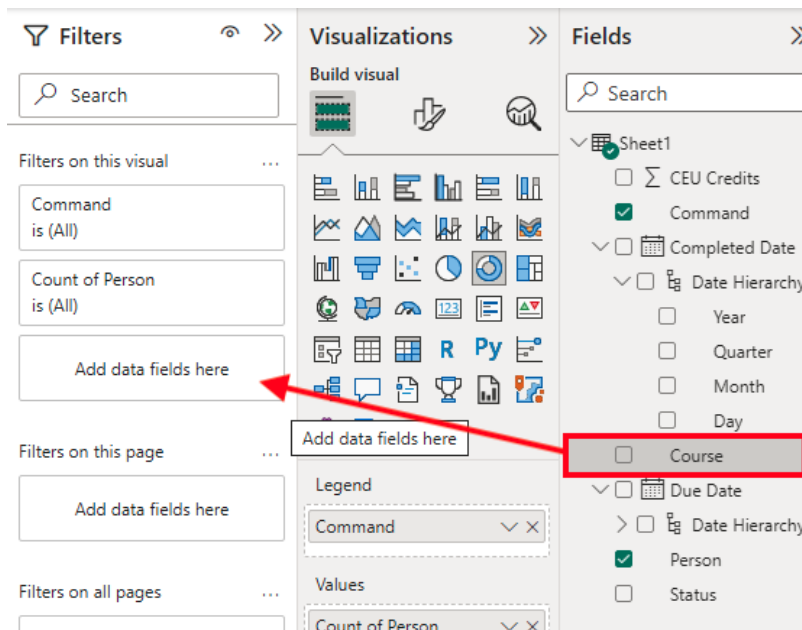
3. Under **Data**, click on the word **PERSON** and drag it to the **Values** section in the *Visualizations* pane.
Then click and **drag** the word **Command** into the **Legend** section in the *Visualizations* pane.



The Donut chart now shows the number of people by Command.

4. Expand out the **Filters** pane

5. Click and **drag** the **Course** field from the *Data* pane to the **Add data fields here** area under **Filters on this visual**



Back in the Filters pane, under the **Course** filter, select **Security+**

Finally, click and drag the **Completed Date** field from the *Data* pane to the **Add data fields here** area under **Filters on this visual**

Expand the **Completed Date** filter. Under **Filter Type** ensure **Basic filtering** is selected, then select **(Blank)**

Filters

Search

Count of Person
is (All)

Completed Date
is (Blank)

Filter type ⓘ
Basic filtering

☒ Select all

☒ (Blank) 8

☐ Wednesday, June 15,... 1

☐ Wednesday, October... 4

☐ Thursday, October 0... 1

☐ Tuesday, November ... 1

☐ Saturday, November... 1

☐ Sunday, November 1... 1

☐ Require single selection

Course
is Security+

Filter type ⓘ
Basic filtering

Search

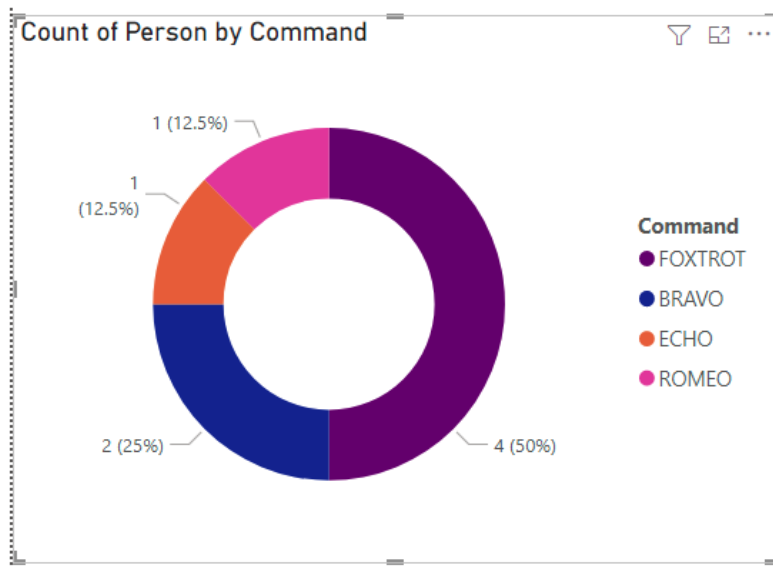
☒ Select all

☐ Cyber Training 13

☐ Security Foundations 17

☒ Security+ 8

You should now have a Donut chart with the number of people who have not completed the Security+ training.



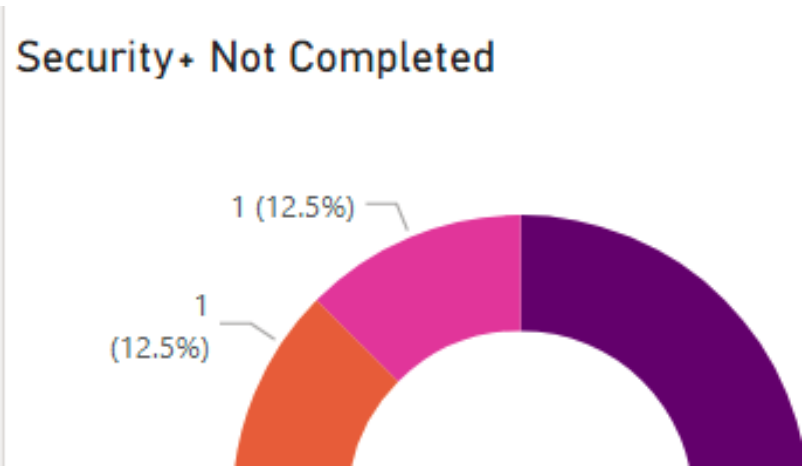
6. Before we move on to the next visual, let's give this visual a proper name. Under **Format Visual**, click on **General**.

Then click **the arrow (>)** next to **Title**

The screenshot shows the "Format visual" pane in Power BI. The "General" tab is selected. The "Title" section is highlighted with a red box. It contains a toggle switch labeled "On" and a text input field with the placeholder text "ity+ Not Completed" and a formula icon.

Under **Text** change that to something more meaningful like "**Security + Not Completed**"

Your chart should now have a nice, descriptive title.

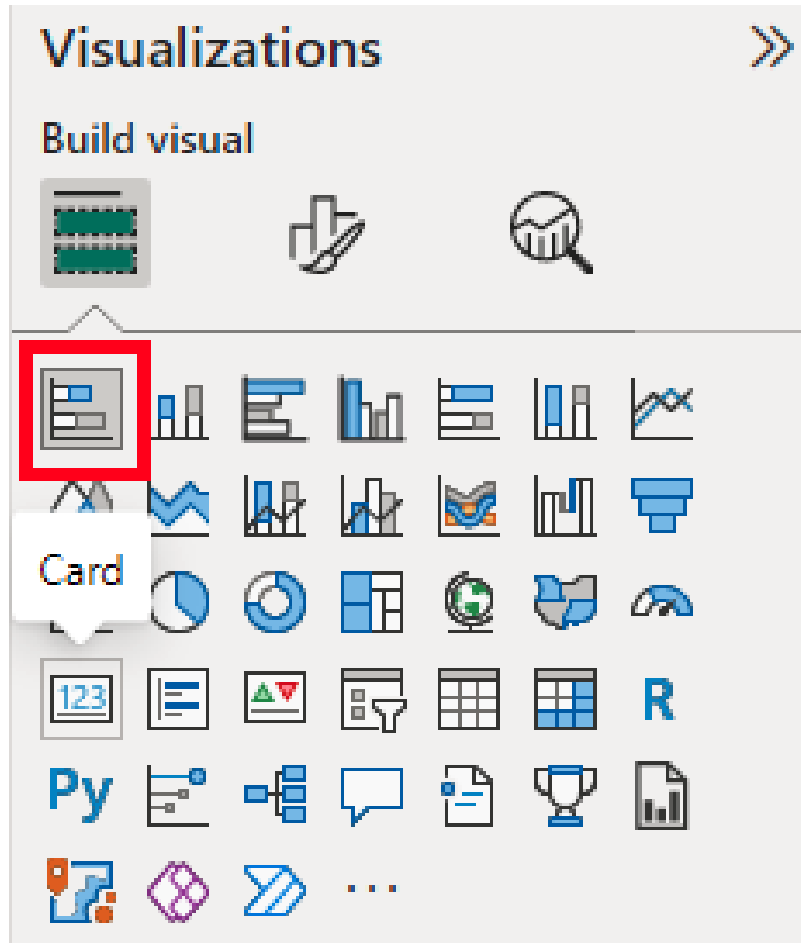


Layout your visuals: Bar Chart

The commander is offering an extra week of leave to the top 5 people who finish their Cyber Training the fastest. We need a quick visual to see who is leading that race.

1. Click on a blank area of the report

Under *Visualizations*, select the **Bar Chart** visual



2. The Y-Axis which is the part that goes up, should display the person's name.

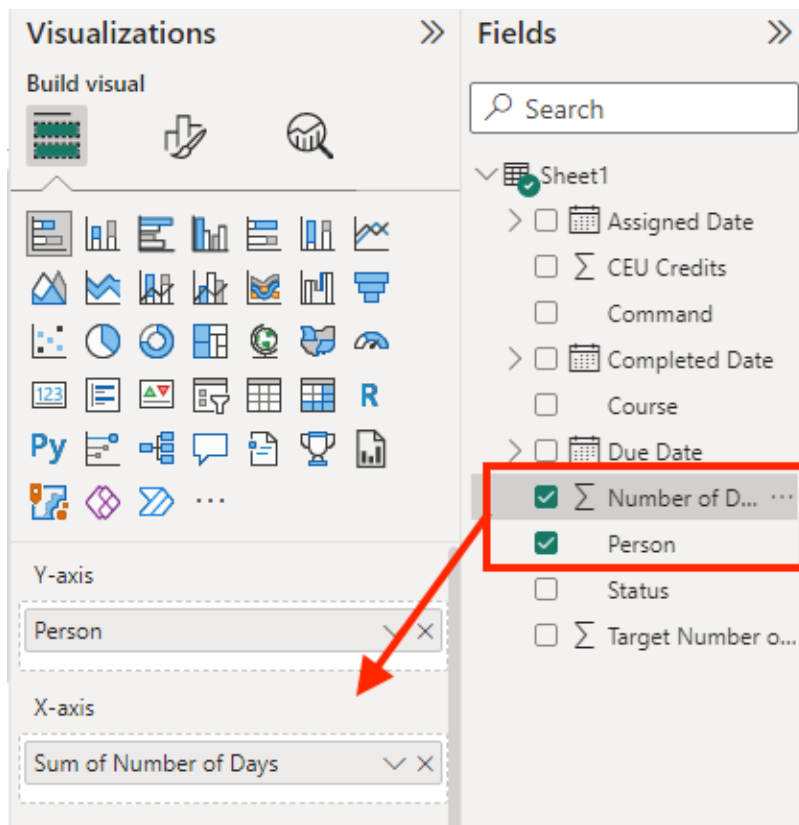
With the new **Bar Chart** visual selected in the report, Under **Data**, click on **Person**.

Person should now be populated under Y-Axis in the *Visualizations* section.

We want the number of days they completed their training to be on the X-Axis, which is the part that goes across.

With the new **Bar Chart** visual selected in the report, Under **Data**, click on **Number of Days** and **DRAG** it to the **X-Axis** field under *Visualizations*.

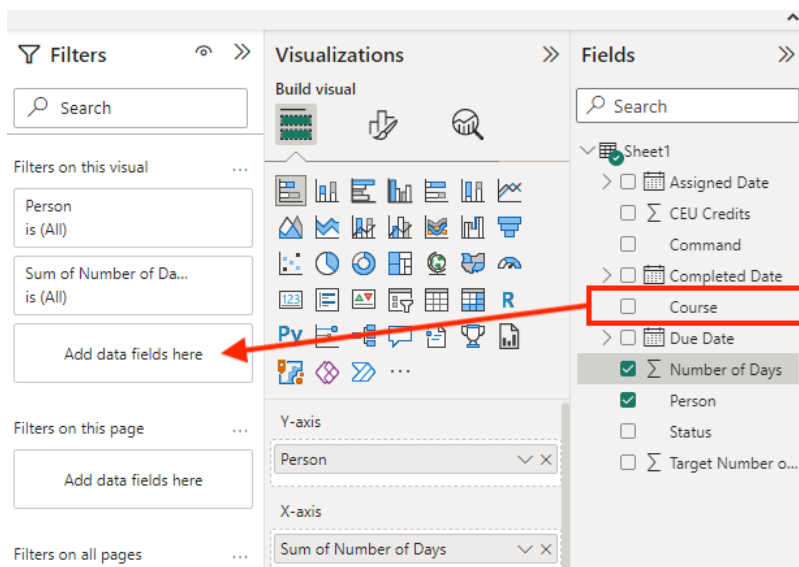
Sum of Number of Days should now be populated under X-Axis in the *Visualizations* section.



3. We want to filter out the other courses and just display the Cyber Training course.

If not already visible, expand out the **Filters** pane.

Under **Data** Click on **Course** and **DRAG** it to the **Add Data fields here** area under **Filters on this visual**,



In the **Course** filter pane, choose **Cyber Training**

4. Because we want the Top 5 people who have finished the fastest, we need to change the *Person* filter.

Expand the **Person** filter.

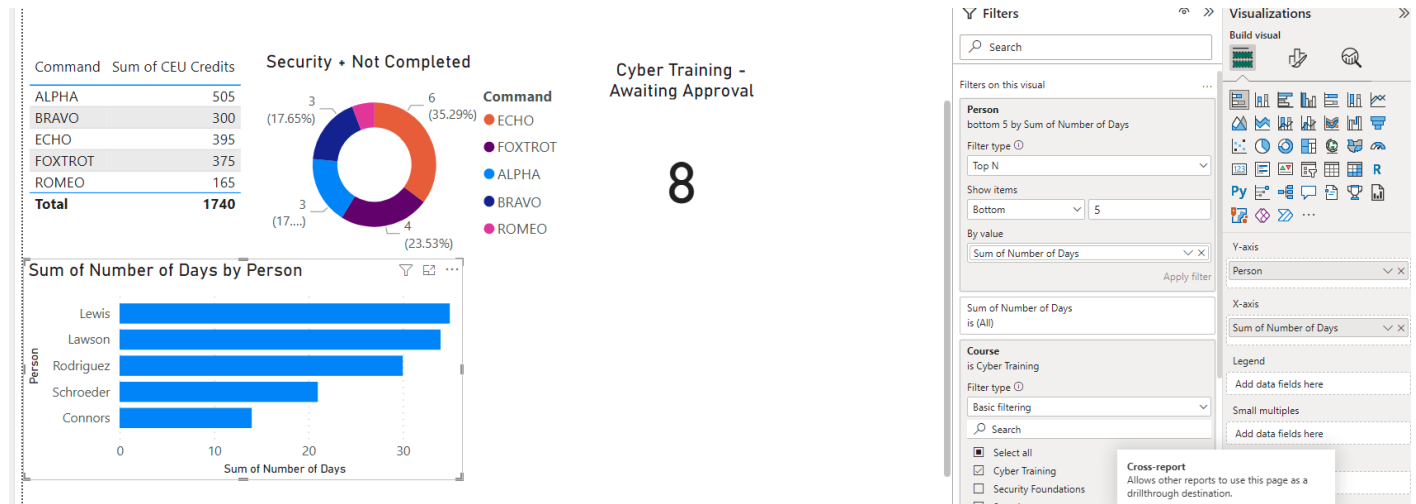
Under **Filter Type** change it to **Top N**

Under **Show items** change *Top* to **Bottom** (We want the smaller numbers on top), then type in the number **5**

Under **By value** click and **drag** the **Number of Days** field to this filter. The value should change to *Sum of Number of Days*

Click **Apply Filter**

You should now have a basic Bar Chart showing the top 5 people who finished the course the fastest, based on the Number of Days column in your spreadsheet.

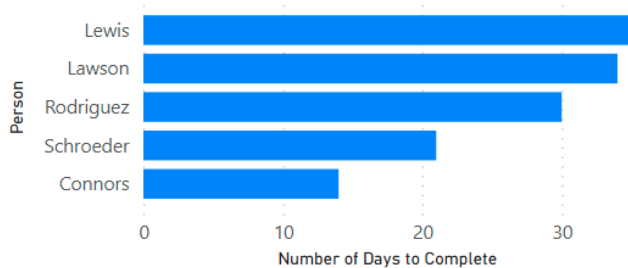


5. Using the same method you used in the last visual,

Change the **Title** to **Commander's Cyber Challenge**

Change the **X-Axis** label to **Number of days to complete**

Commander's Cyber Challenge

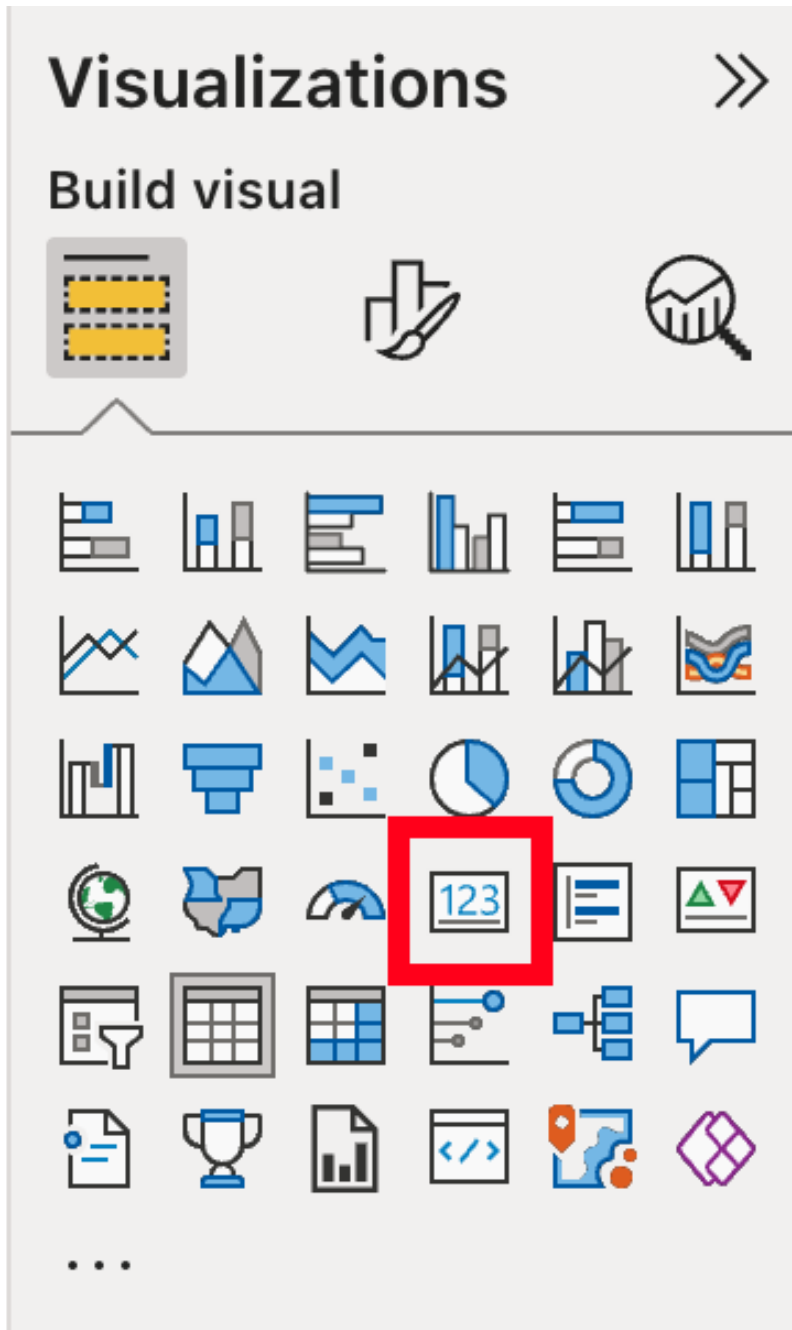


Layout your visuals: Card Visual

Sometimes just a simple number or statistic is all you need to encourage action from the consumer of this report. Let's add a simple card that shows the number of personnel that is in *Awaiting Approval* status for a specific course.

1. Click on a blank area of the report

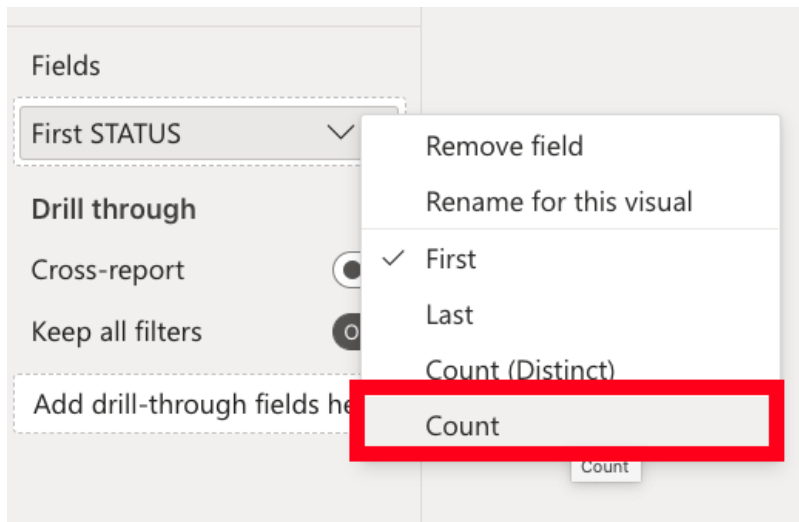
Under *Visualizations*, select the **Card** visual



2. With the new **Card** visual selected in the report, Under **Data**, click on **STATUS**.

First STATUS should now be populated under *Fields* in the *Visualizations* section.

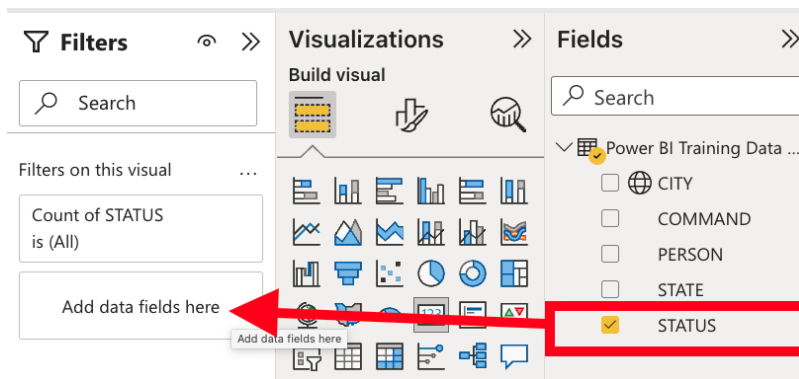
Under the **Visualizations** tab under **Fields**, Click on the **down arrow (V)** next to **First STATUS** and change it to **Count**.



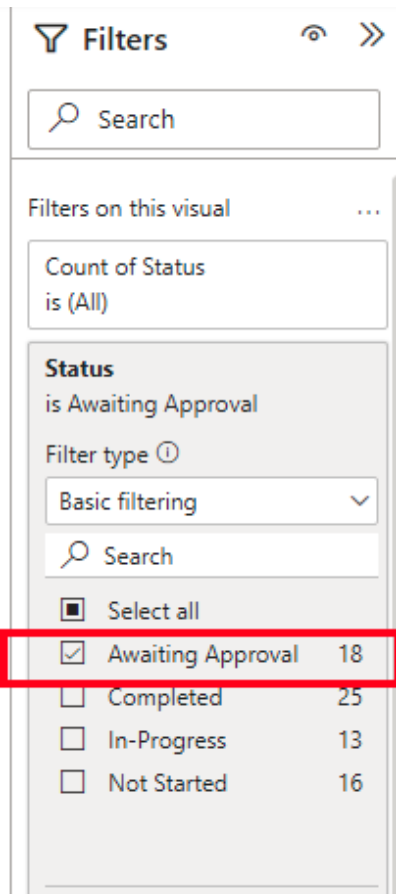
For this visual to display the number of personnel who are in *Awaiting Approval* status, we will need to add the *STATUS* Filter to the visual.

1. If not already visible, expand out the **Filters** pane.

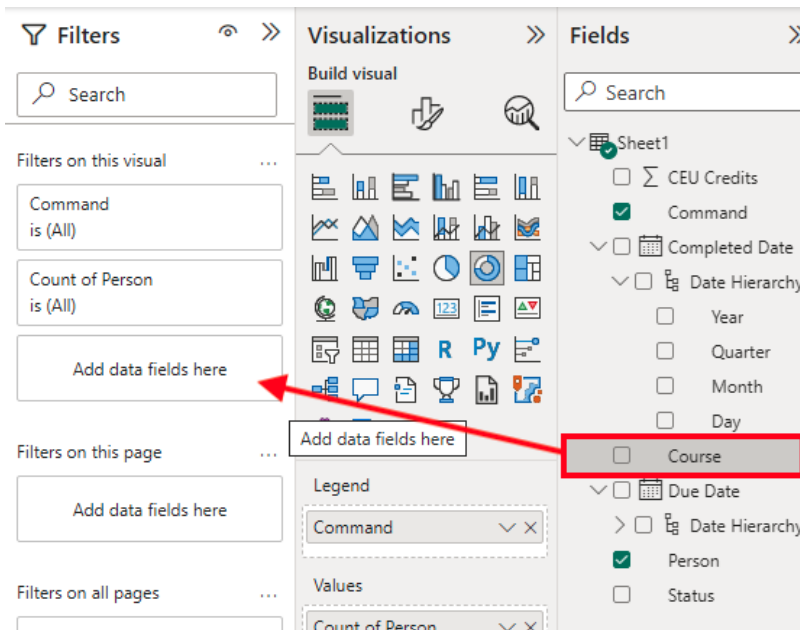
Under **Data** Click on **STATUS** and **DRAG** it to the **Add Data fields here** area under **Filters on this visual**,



2. In the **Status** filter pane, under *Basic Filtering*, select **Awaiting Approval**



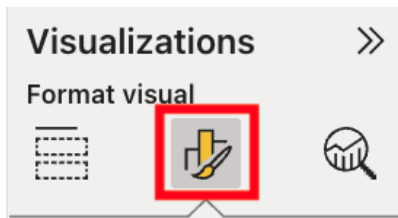
3. Under **Data** Click on **Course** and **DRAG** it to the **Add Data fields here** area under **Filters on this visual**,



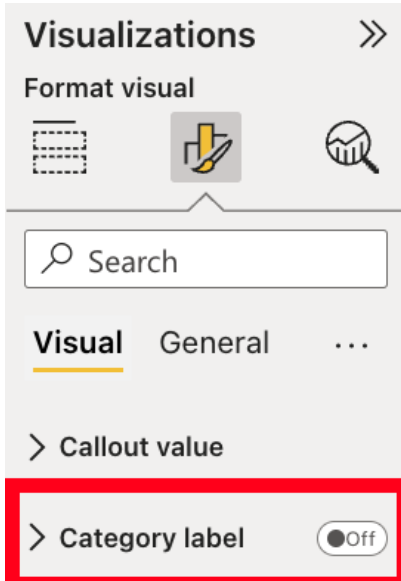
4. In the **Course** filter pane, choose **Cyber Training**

This is great, but it would be easier to understand with a few tweaks to describe what this visual is telling us.

1. Click and select your card, then under *Visualizations*, select the **Format Visual** icon (in the middle).

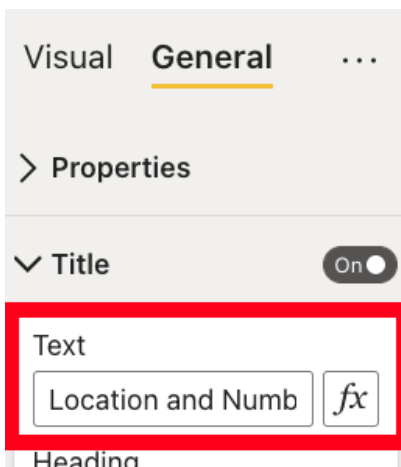


2. Under **Visual** Next to **Category labels** toggle the button to **Off**



3. While we are here, let's give this visual a proper name. Under **Format Visual**, click on **General**.

Then toggle the selector to **On**

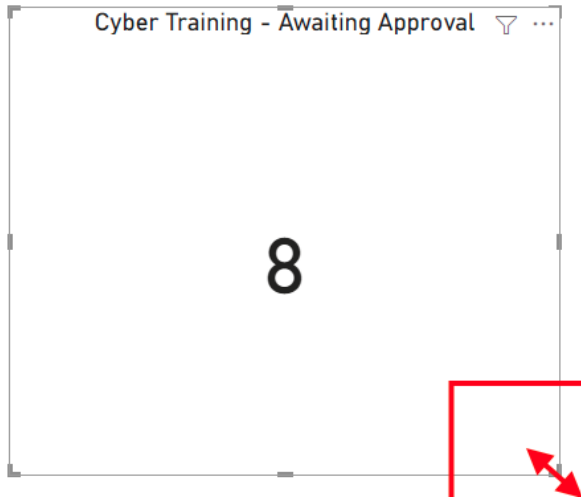


Under **Text** change that to something more meaningful like "**Cyber Training - Awaiting Approval**"

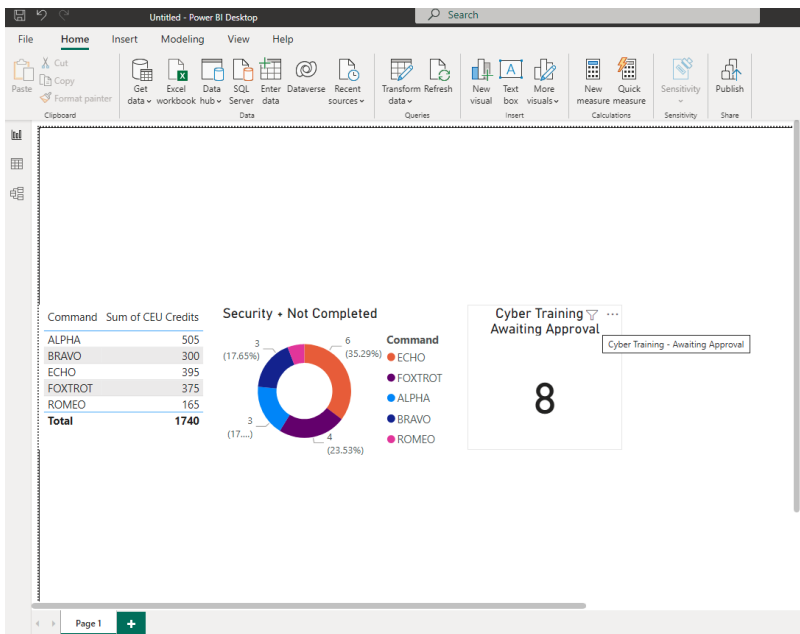
Your card should now have a nice, descriptive title and a sample number.

The card visual is just WAY TOO BIG! Let's resize it so that we can fit one more visual on this report.

1. **Hover your mouse** over the** right corner bracket** of the *Cyber Training* card until it changes to an arrow.



2. **Click and hold** while resizing the card to approximately the same size as the other cards. Use your judgment to determine what looks best.

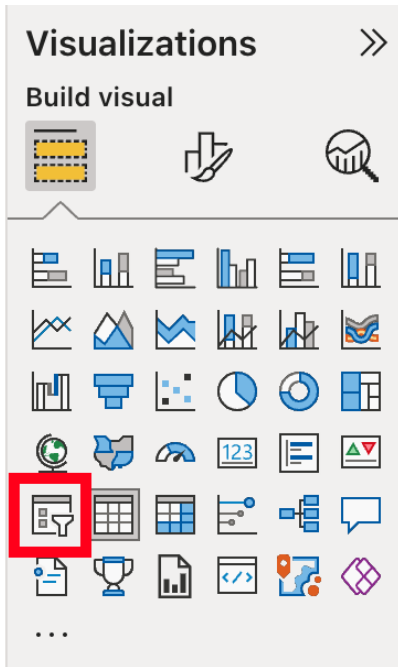


Layout your visuals: Slicers

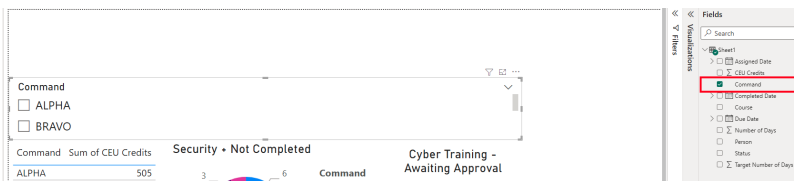
Since this will be used by the Command Staff, we want to make it as easy as possible to filter out the data they don't want to see. To do this, we'll create a *Slicer*.

1. Click on a blank area of the report

Under *Visualizations*, select the **Slicer** visual icon



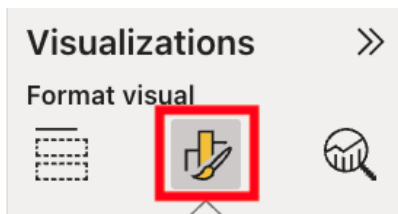
2. With the new **Slicer** visual selected in the report in the report, Under **Data**, click on **COMMAND**.



Your Slicer should now have a selectable list of *COMMANDS*

This is functional, but let's make this a little bit prettier and easier to use.

3. Click and select your Slicer, then under *Visualizations*, select the **Format Visual** icon (in the middle).

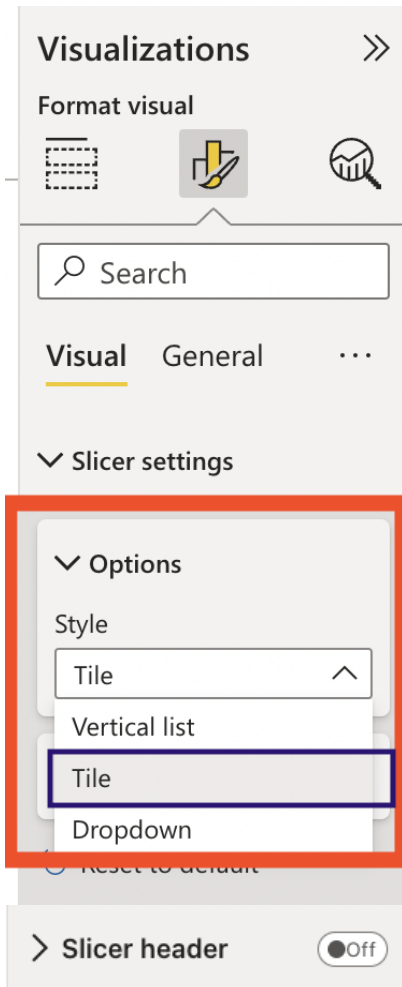


Feel free to resize your slicer frame as you wish. Our example has it in the middle, leaving room on top for a title and logo (which we will do next)

4. Under **Visual** click the arrow (>) next to **Slicer Settings** then again next to **Options**

5. In the **Options** field, change it to **Tile**

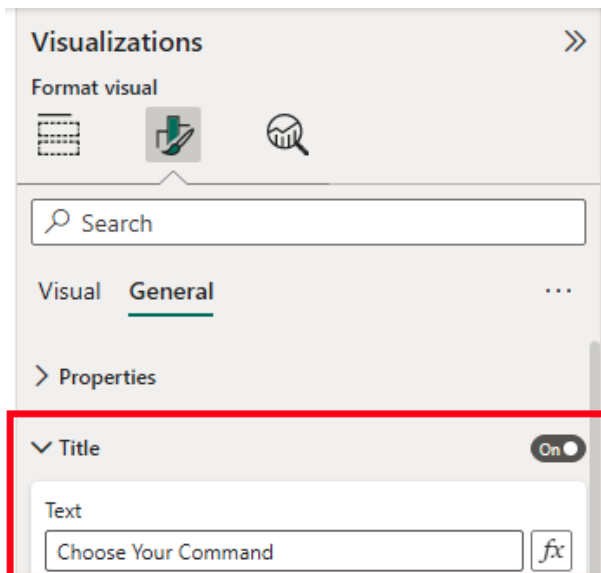
Then toggle the **Slicer Header** to **Off**



While we are here, let's give this visual a proper name. Under **Format Visual**, click on **General**.

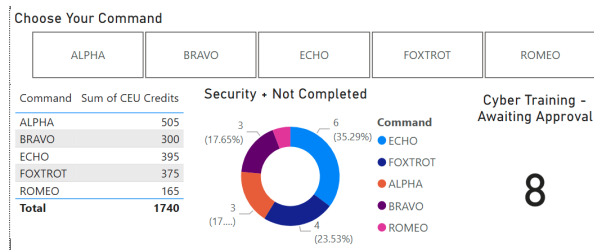
6. Click the toggle next to **Title** to **On**

Then click **the arrow (>)** next to **Title**

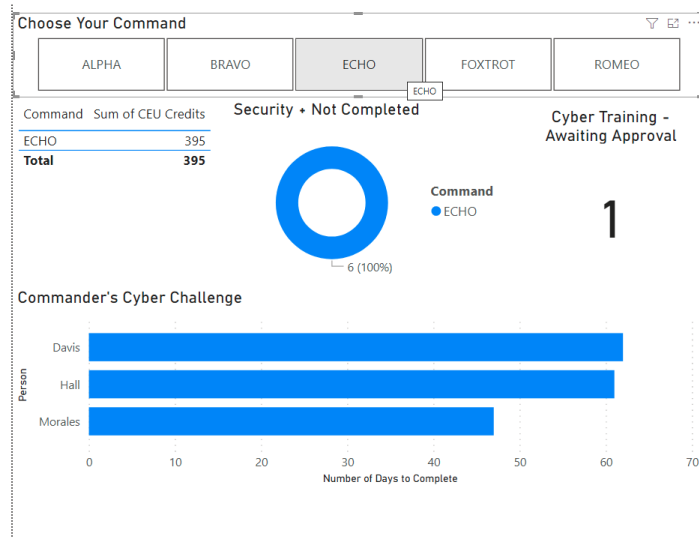


Under **Text** change that to something more meaningful like "**Choose your COMMAND**"

Your Slicer should now have a nice, descriptive title and larger buttons.

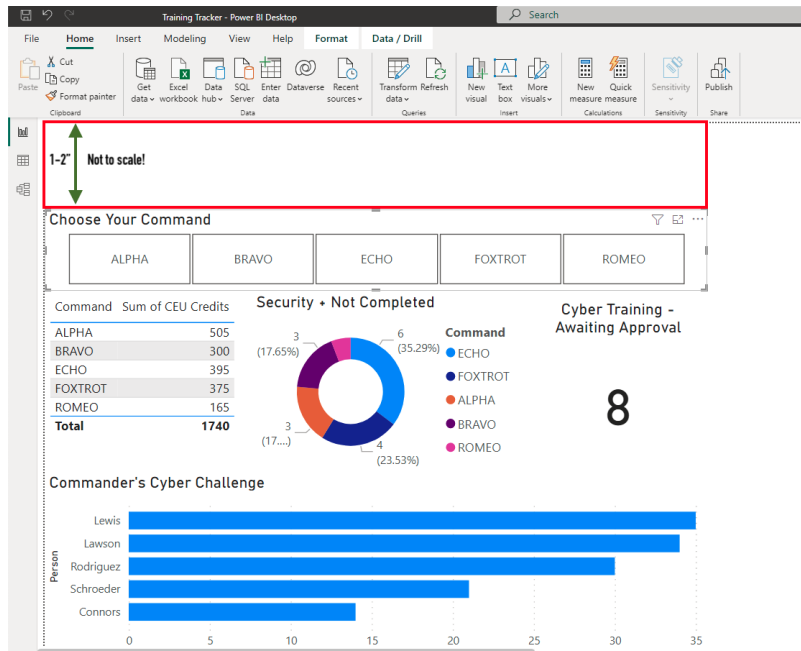


Go ahead and click on one of the COMMANDS in your Slicer. You will see the values of your report visuals change.



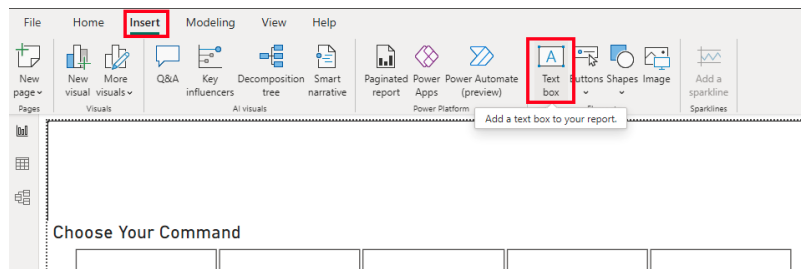
Add a report title text label

We need to make some room on top for our report title. You can click and drag your visuals around the report to make space for the title on the top. Feel free to resize and reposition them as well to suit your preferences. You should have about an inch or so at the top of your report for the next step.



1. On the top menu bar on the right side, click on the **Insert** menu.

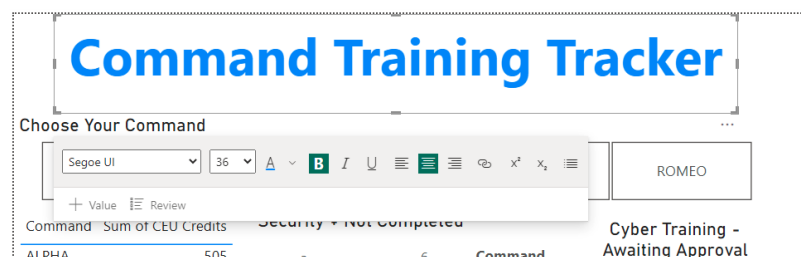
Click on **Text Box**



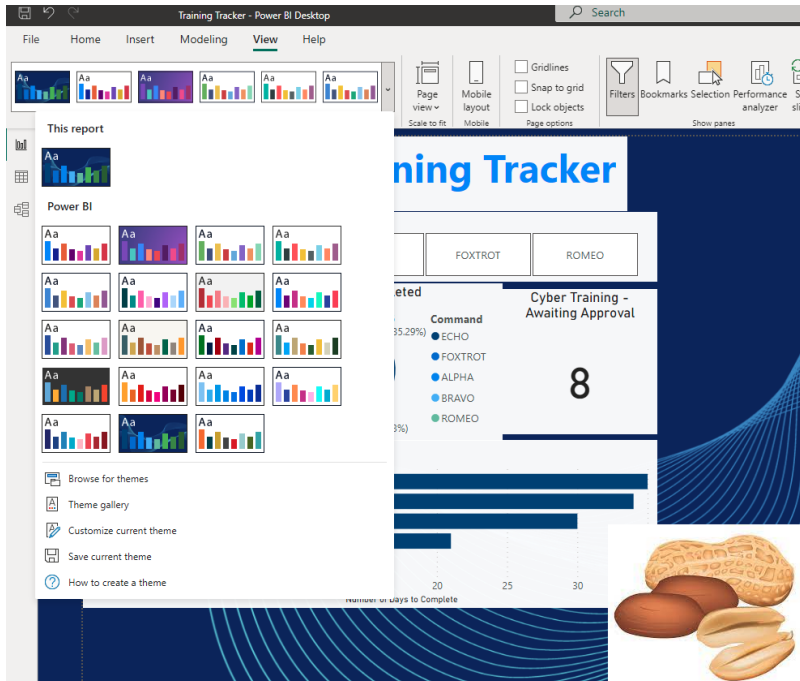
2. Move and resize your Text Box to the top center of your report (or wherever you like). Don't forget that you can use the guide markers to help you center your title.

Inside the **text box**, click and type in a title for your report. Something like "**Command Training Tracker**".

In the Text Label options box, select the **Font**, **Size**, and other formatting options for your title. We recommend making it big and easy to read.



Feel free at this time to customize your report as you wish. Experiment with new themes, borders, shadows, and colors. Resize and move your visuals around. Go nuts! :)

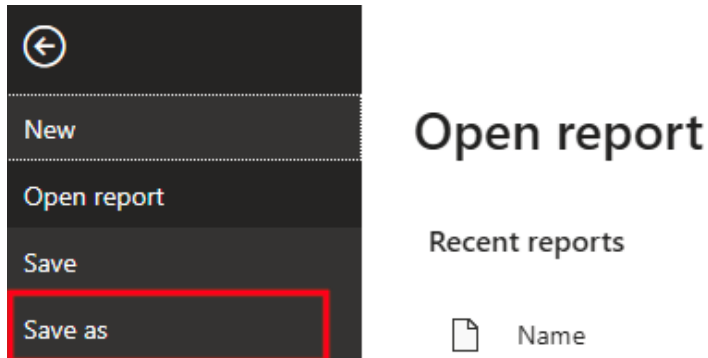


Save your Report

You don't want your hard work to go to waste! Let's get this saved so that you can publish your work or edit it later on.

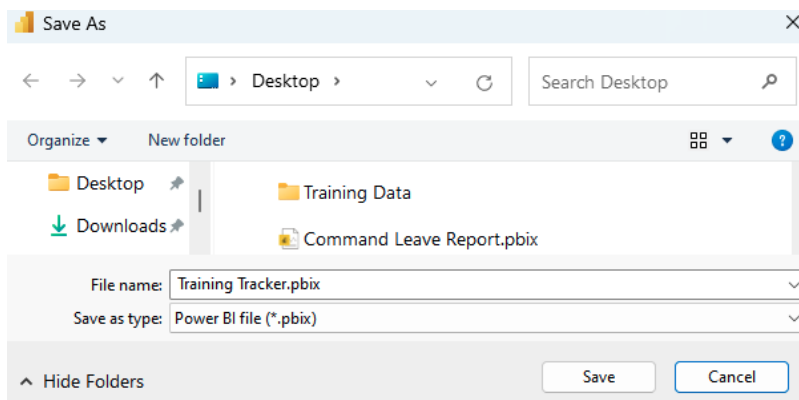
1. Click on the **File Menu**

Click **Save As**



Give it a meaningful name and location where you can find it easily (you'll need it for the next Activity)

Click **Save**



You've now created a very simple Power BI Desktop Report using some simple visuals. From here you can build upon what you have learned into much more advanced reports and visuals. Power BI Desktop is a very powerful tool that can help you truly visualize your data!

THIS COMPLETES ACTIVITY #1